Examiner only

Answer **all** *questions.*

1. *(a)* The diagram below shows **part** of the electromagnetic (em) spectrum.

Microwaves	Infra-red	Visible light	Ultraviolet

Use **only** the regions of the em spectrum **shown in the diagram** to answer the following questions.

(i) Name the region of the em spectrum with the longest wavelength. [1]

....

(ii) Name the region of the em spectrum with the lowest frequency. [1]

....

(b) Name **one** region of the em spectrum not shown in the diagram in part *(a)*. [1]

....

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(c) Waves can either be described as transverse or longitudinal. Sound waves are an example of longitudinal waves whereas visible light waves are transverse.
Tick (✓) the **two** correct statements below. [2]

Ultraviolet waves are longitudinal waves ☐

Longitudinal waves cannot be reflected ☐

Microwaves are transverse waves ☐

In a longitudinal wave the vibration of the particles is parallel to the direction of the wave ☐

Sound waves travel slowly in a vacuum ☐

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(d) The table below gives information about the frequency and wavelength of sound waves in different materials.

Material	Frequency (Hz)	Wavelength (m)
air	170	2
water	170	9
iron	170	29

Use the information in the table to answer the questions below.

(i) Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the speed of sound waves in air. [2]

Wave speed = ... m/s

(ii) The sound wave travels from air into water. Its frequency stays the same. Without further calculation explain whether its speed increases, decreases or stays the same. [2]

....

....

....

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